
Clockworks™
Automated Insights for Smarter Facilities
CLOCKWORKS
ANALYTICS

Facilities organizations face expanded energy, operational efficiency, and reliability goals while aspiring to realize visions for analytics, IoT, and remote management under constrained budgets and staffing. Technology can help.



80%

Of equipment fail for non-age-related reasons, such as insufficient maintenance

40%

Of employees report uncomfortable conditions, too hot, too cold, blowing air...

20%

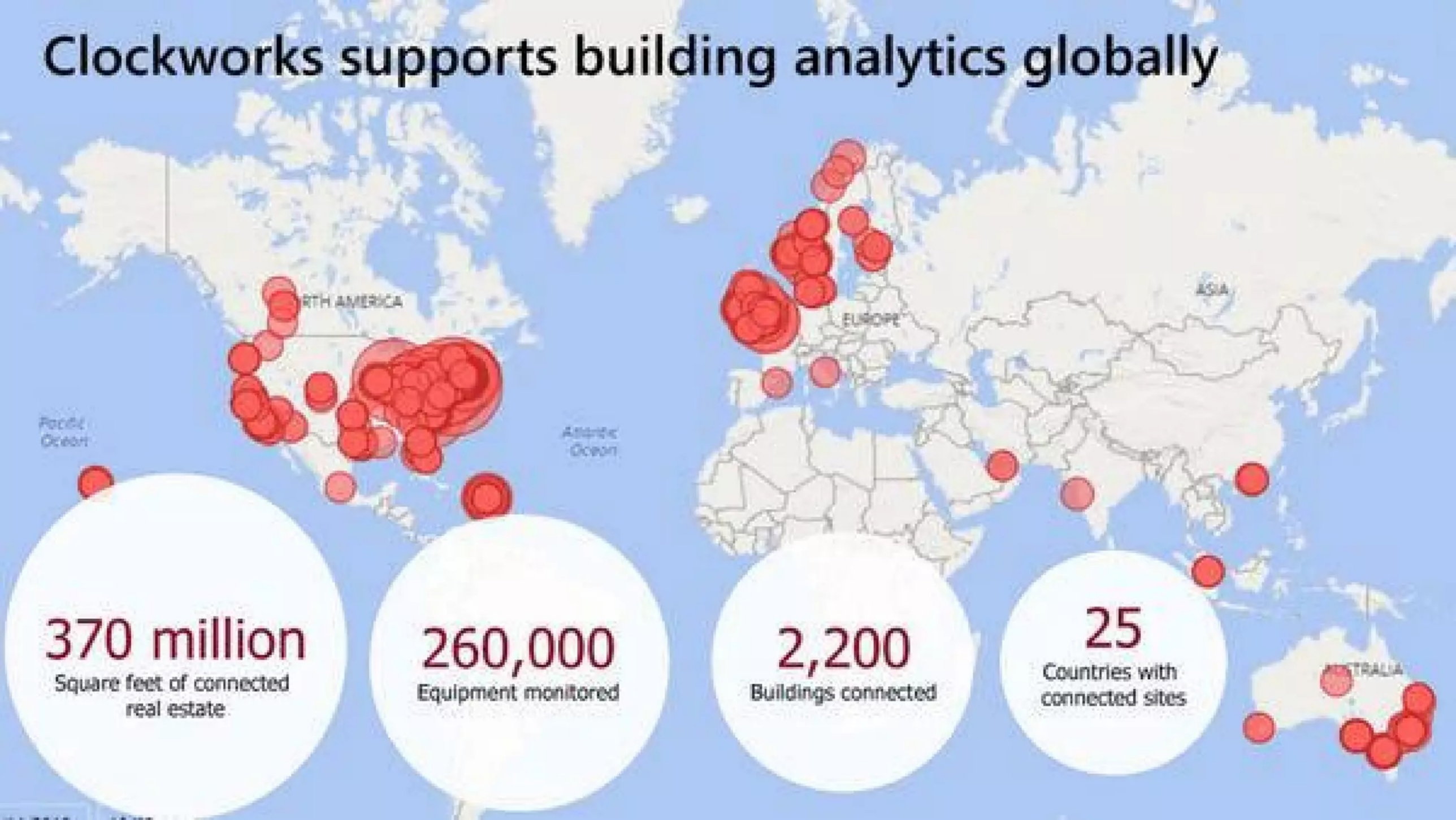
Energy and operational expenses could be saved using predictive maintenance

CLOCKWORKS
ANALYTICS

We empower facilities teams to run smarter buildings with data-driven proactive maintenance



Clockworks supports building analytics globally



370 million
Square feet of connected real estate

260,000
Equipment monitored

2,200
Buildings connected

25
Countries with connected sites

Our Customers were awarded all of the DOE FDD awards in 2019



The Big Picture – How Facilities Leaders are Realizing Benefits

Reduce energy cost & achieve sustainability goals	Reduce energy cost, energy consumption, and carbon impact by finding and fixing issues otherwise unseen
Increase operational efficiency	Allocate maintenance resources towards prioritized issues to achieve cost and resource efficiency
Improve system reliability	Proactive, condition-based maintenance to extend equipment life, improve comfort, and manage risk

Client Results: Energy Prioritization Reducing Energy Cost and Consumption

Completed Tasks

1114

Annual Projected Task Savings

$1,640,683

With Comfort Impact

472

With Maintenance Impact

922

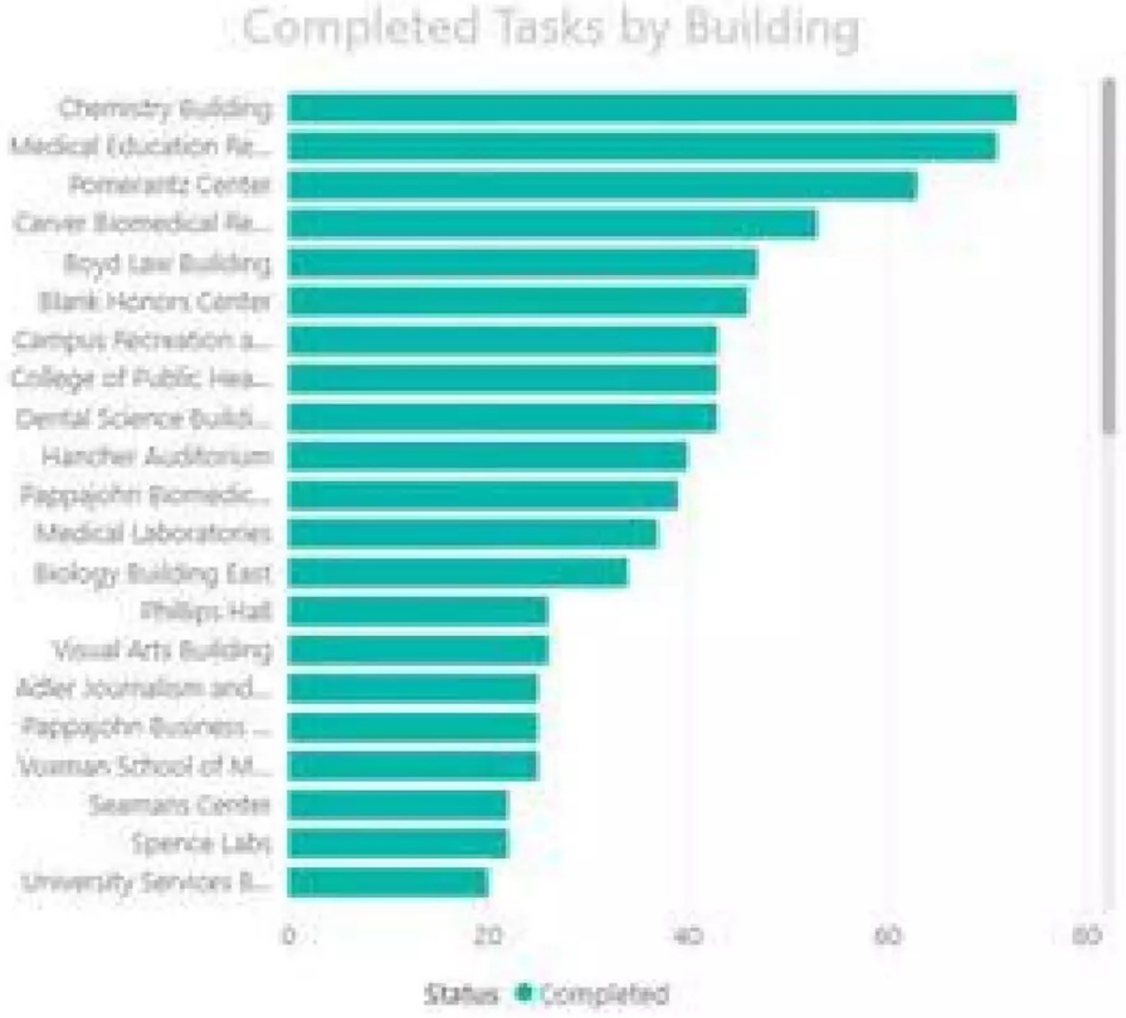
With Energy Impact

249

With calculable savings

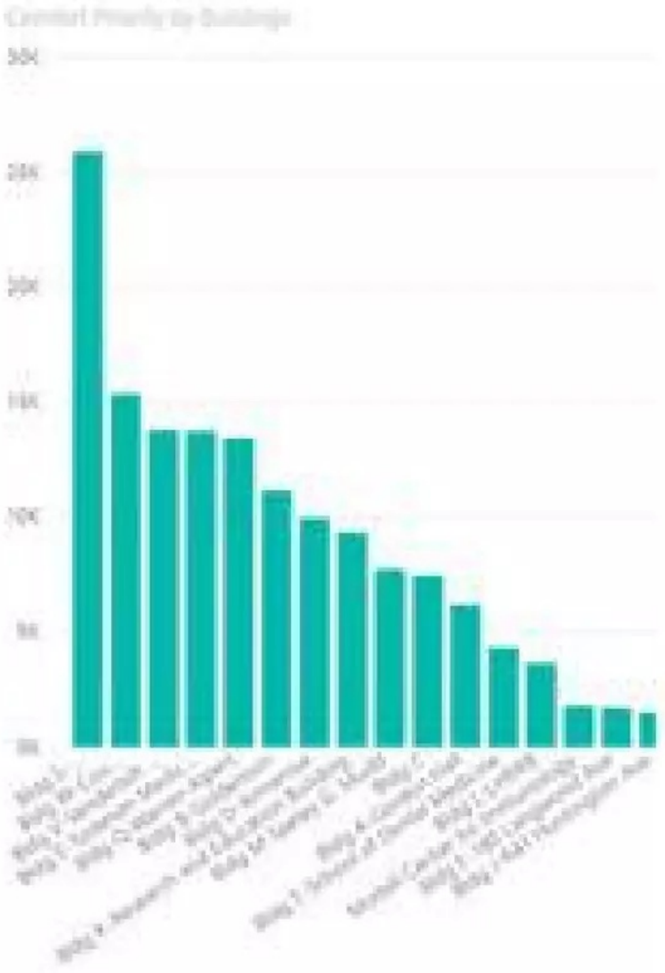
205

With non-calculable savings



Fault prevalence informing maintenance priorities and capital budgets

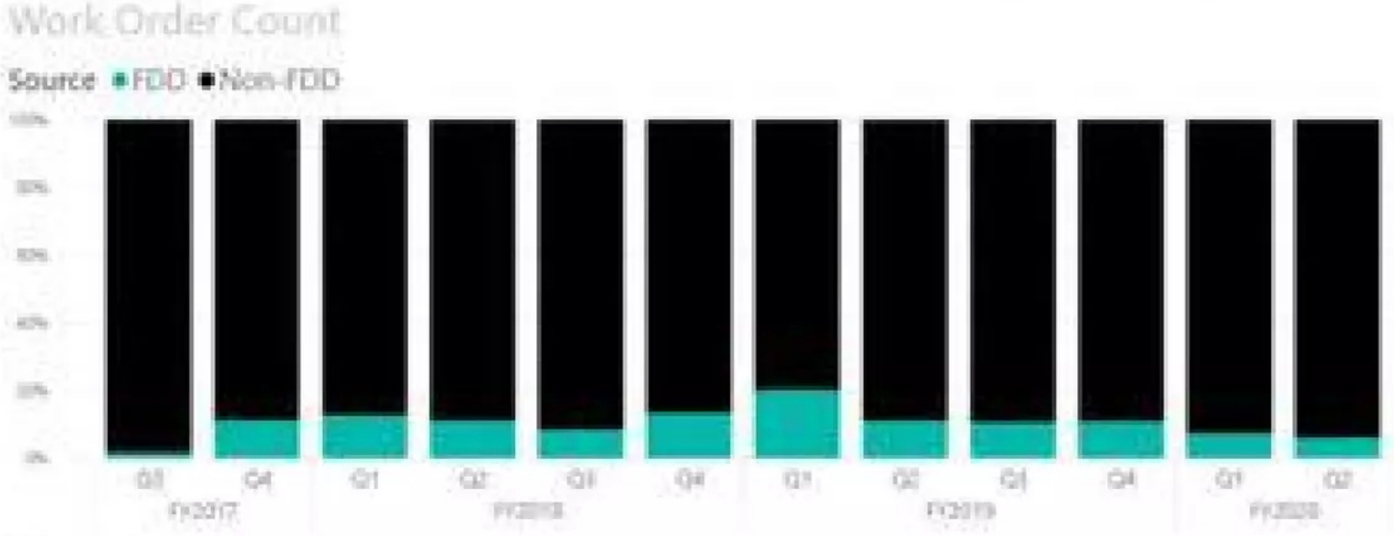
Buildings	Comfort Priority	First Da...
Bldg S-Flexi- Research Building	20000	%
Bldg W-Countway Library	19317	%
Bldg V-Vanderbilt Hall	13799	%
Bldg B-Thorndike Medical Education Center	13729	%
Bldg Q-Warren Alpert	13400	%
Bldg E-Goldenson	11151	%
Bldg D-Amerinia	9982	%
Bldg K-Research and Education Building	9323	%
Bldg M-Seeley G. Mudd	7758	%
Bldg C	7453	%
Bldg A-Gordon Hall	6170	%
Bldg T-School of Dental Medicine	4319	%
Bldg L-HHRI	3841	%
Model Center for Immunology	1783	%
Bldg F-180 Longwood Ave	1602	%
Bldg J-640 Huntington Ave	1480	%
Total	146900	%

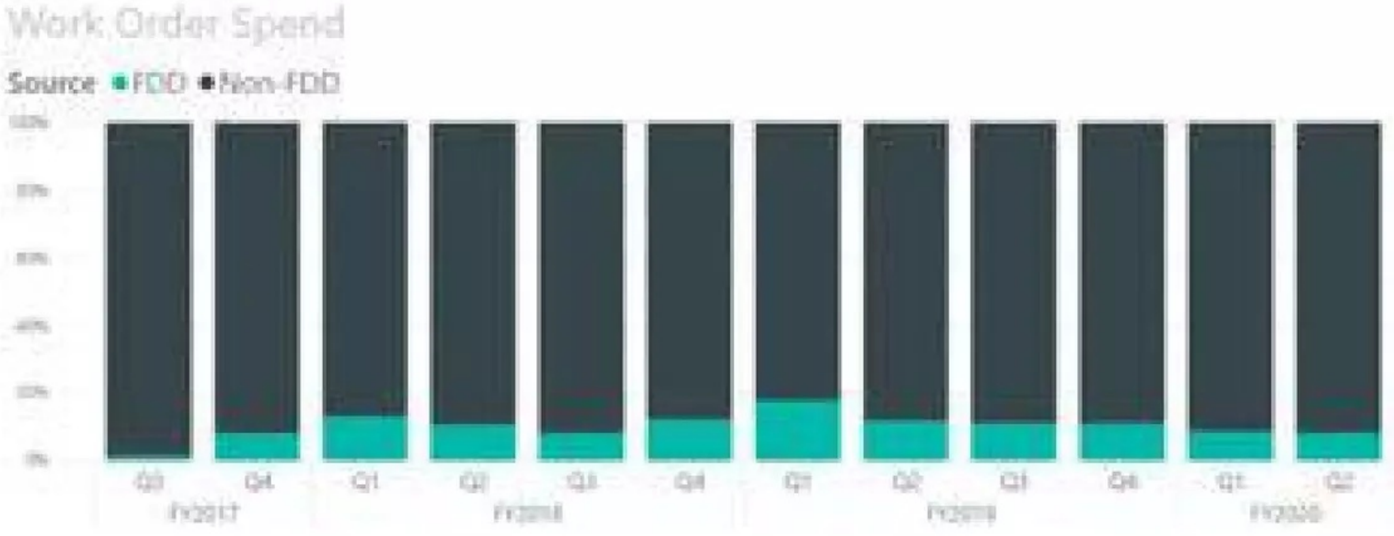


Faults

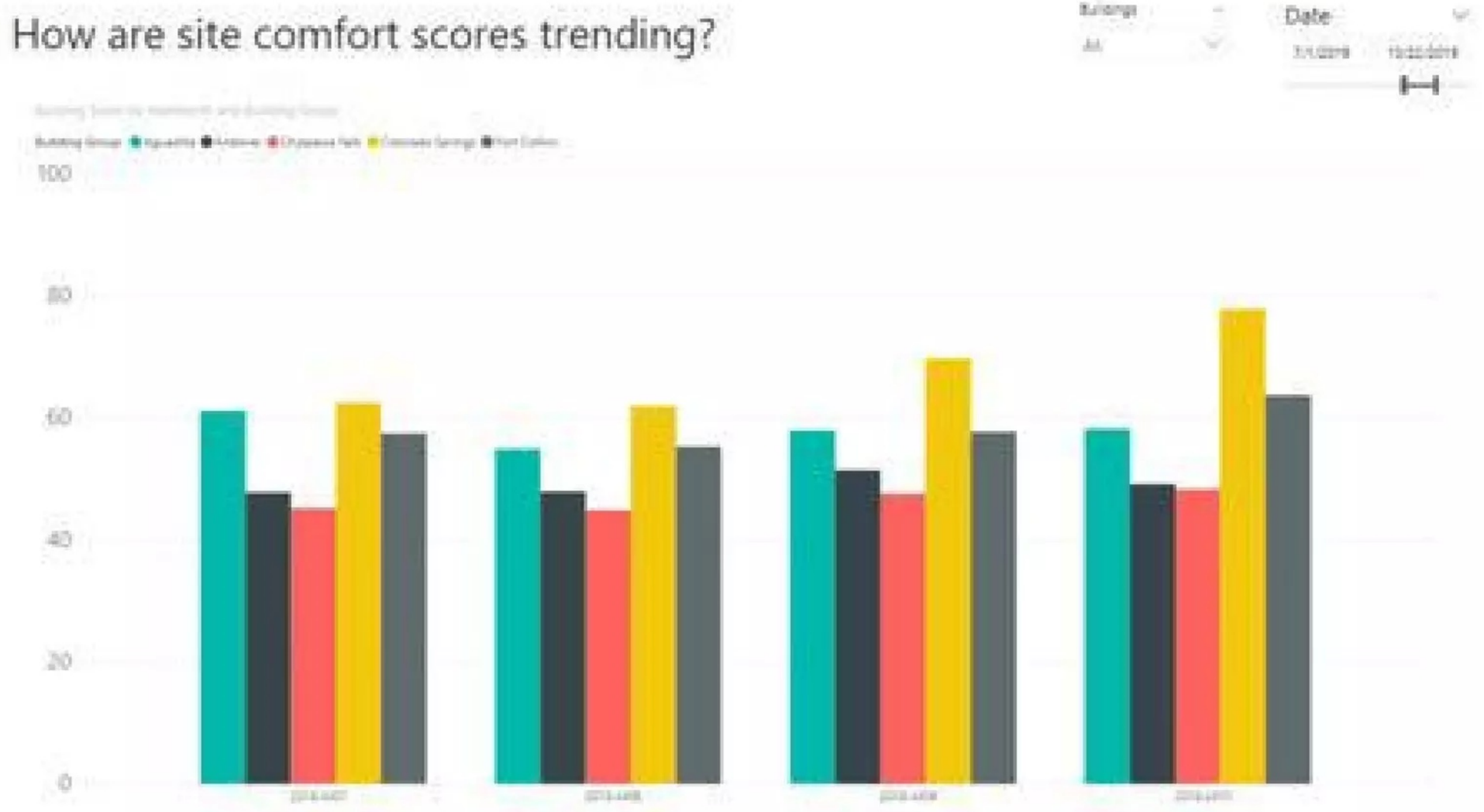
Fault	Faults
Room air temp higher than setpoint	[redacted]
Sensor error (Flat)	[redacted]
Room air temp cooler than setpoint	[redacted]
Zone occupied, AHU off	4791
Supply flow below minimum	[redacted]
Missing information: fan hp, fan flow	[redacted]
Flat sensor error	[redacted]
Stuck cooling valve	[redacted]
Stuck heating valve	[redacted]
Zone quality flag	[redacted]
Sensor error on flow reverse	[redacted]
Room air temp not tracking setpoint	[redacted]
Missing information: AHU fan rated HP/flow	[redacted]
Setpoint was zero	[redacted]
Supply fan status data mismatch	[redacted]
Heating and cooling setpoints close together	[redacted]
Zone fan on while unoccupied	[redacted]
Zone supply air temperature short cycling	[redacted]
Stuck zone supply damper	[redacted]
Leaking heating valve	[redacted]
Simultaneous heating and cooling	[redacted]
Fan status data mismatch	[redacted]
Exhaust flow above maximum	[redacted]

Condition-based maintenance reducing reactive and scheduled maintenance





Fault Data creating comfort metrics to hold facilities vendors accountable



Success Story: $1,800,000 savings across five sites at a top-tier pharmaceutical company



A Fortune 500 biotechnology company operating five campuses in three different countries needed automated analytics to drive a smart and integrated facilities program and achieve energy goals.

IMPLEMENTED GLOBALLY

6 global locations

90 buildings

8,1 million square feet

9,900 equipment connected

15 million daily points

TRANSFORMATIVE IMPACT

Centralized operations center serving global portfolio

$1.8M savings in the first two years

$5M savings to date

International award for innovative energy project

Success Story: Fault detection transforms facilities at a global information technology company



A global technology company implemented fault detection to enable predictive maintenance and energy savings using their global facilities management services provider.

NATIONAL ROLLOUT

5 sites

15 buildings

2 million square feet

4,400 equipment connected

8.6 million daily points

Link to success story

FUTURE FACILITIES MANAGEMENT

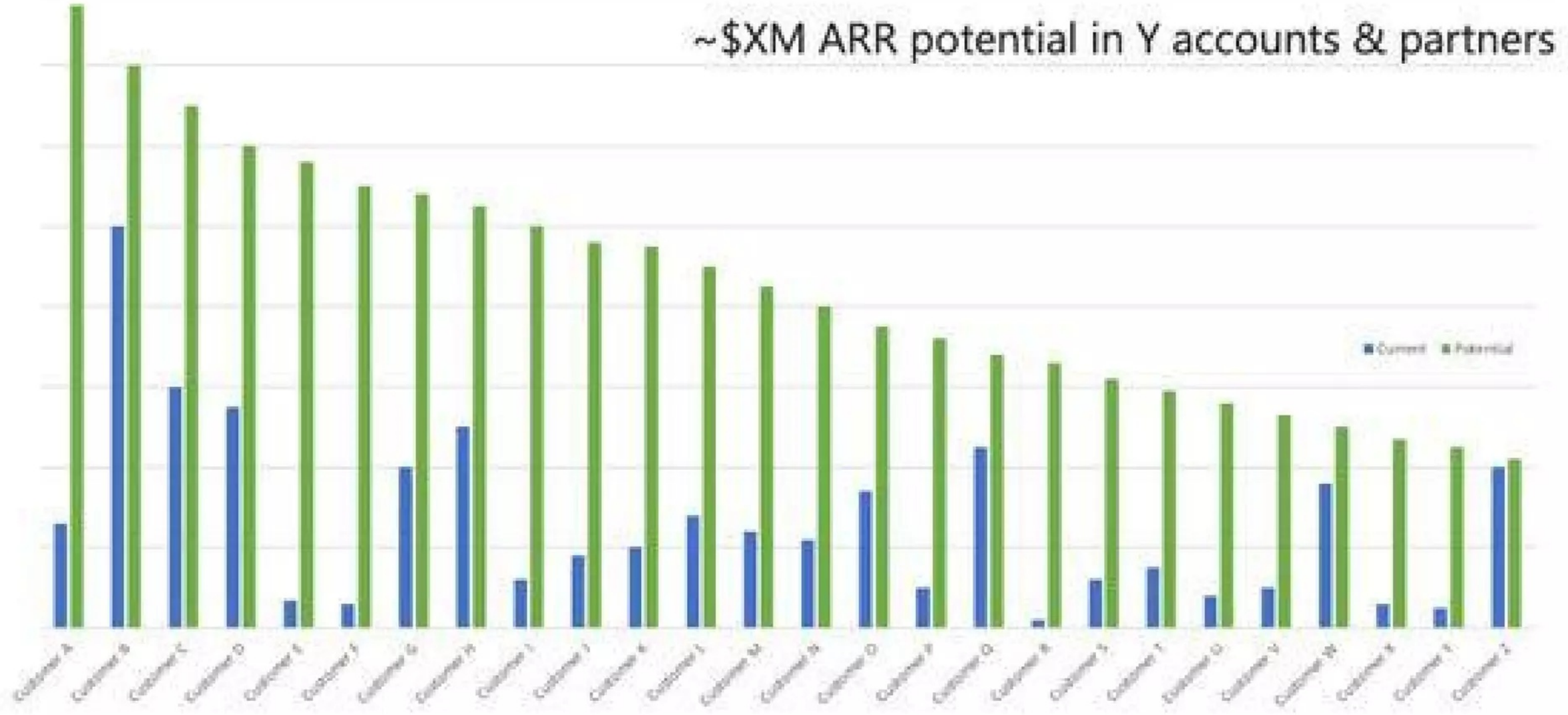
Integrated into facilities management provider service

Reduced hot and cold calls

9% energy savings

Awarded for innovation by the US Department of Energy backed Smart Energy Analytics Campaign

ARR Expansion Potential



The market is big and our momentum is strong

$11B
Building Analytics market emerging according to industry analysts

Redacted
ARR under contract

50%
2018-2019 YoY Revenue growth

$7M
Expansion ARR potential in 27 largest enterprise accounts alone

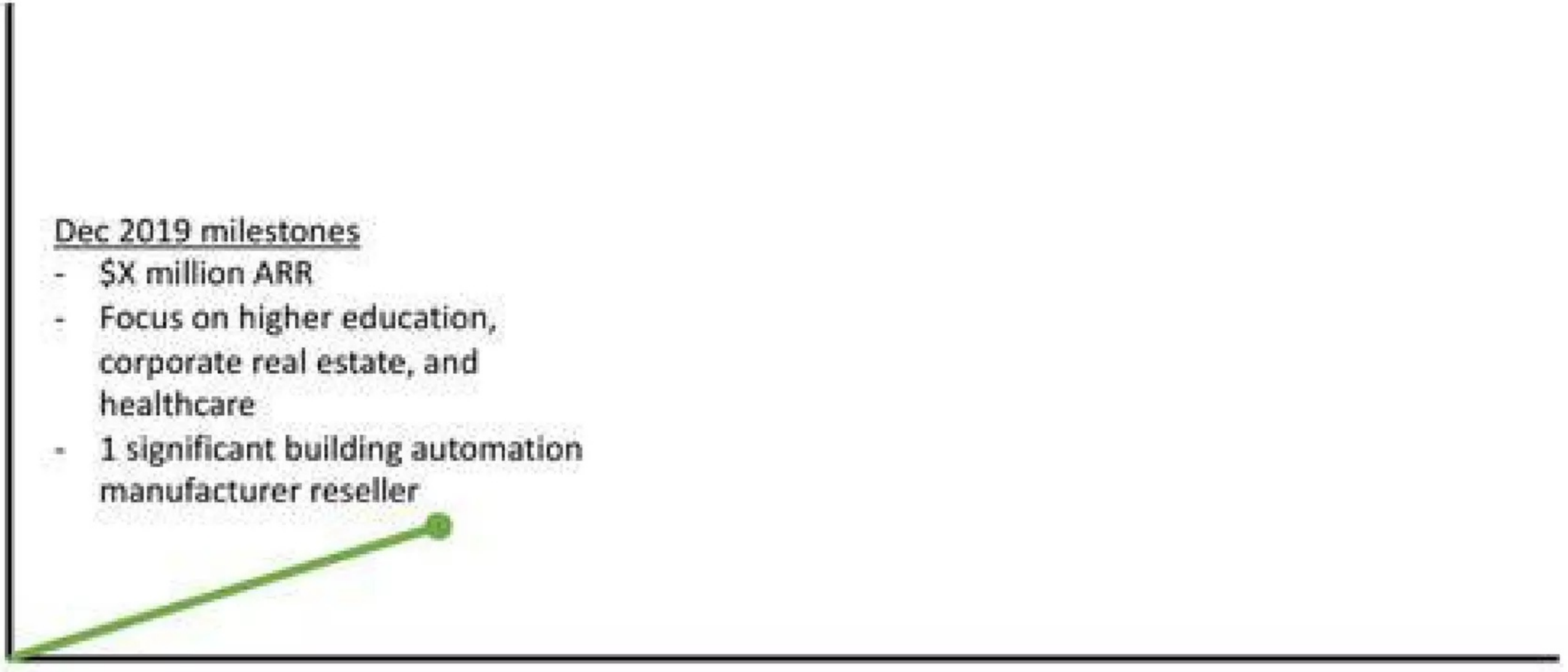
Service Partners Powered by Clockworks

A New Way To Do Business	Include Clockworks in your standard or premier service offering
Value Add	Building Control Services, Mechanical Services, Energy Services, Facility Management, Commissioning
Vision	We work with you to help your customers understand the transformation in facilities processes and the outcomes in creates

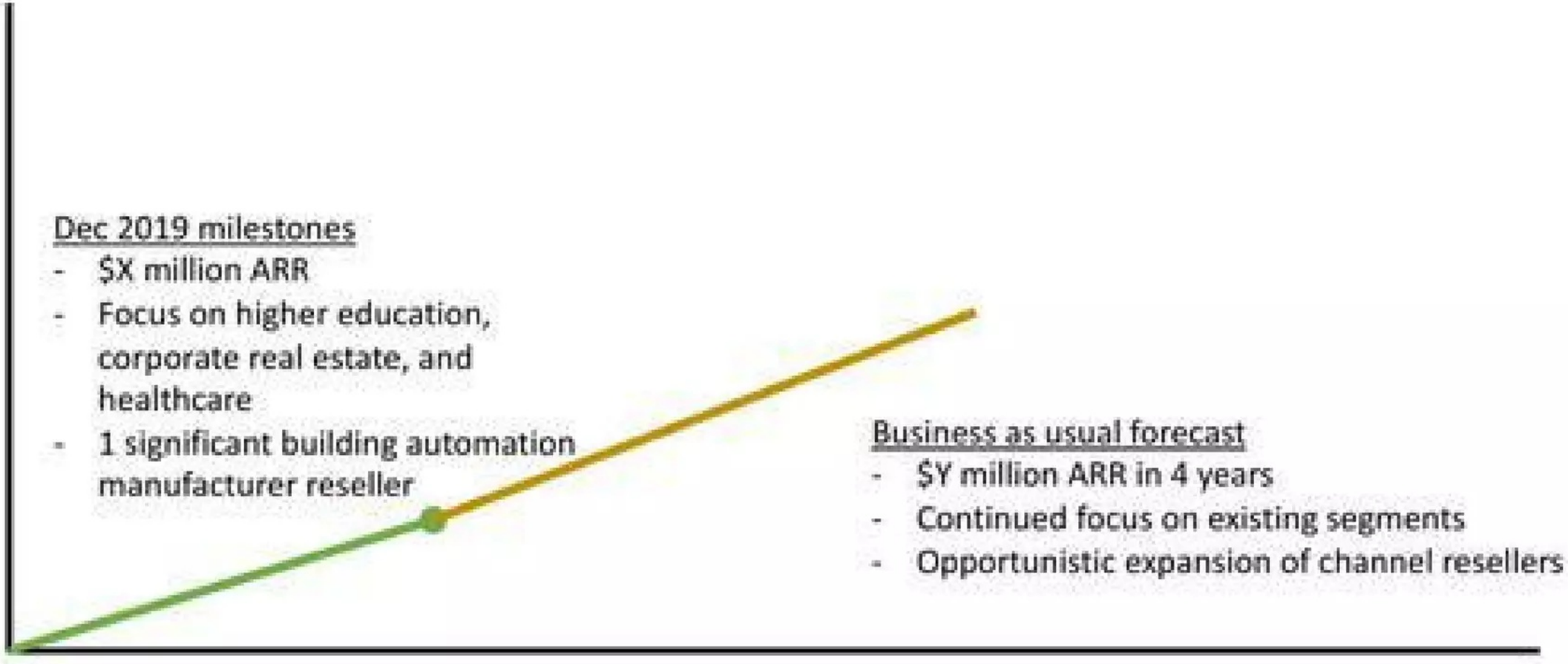
It’s about Transforming Partner Services

Collaboration	Provide guidance, incorporate FDD into business processes to transform facility processes and outcomes.
Training	Monthly service meetings, software support, and engineering support at corporate level
Results	Persistent energy savings, Pro-active condition based-maintenance, efficient building operation, super charged M&V

We seek investment to drive channel growth and new market expansion



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